

Beckham Moreland

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Education

San Diego State University, College of Engineering
B.S. in Aerospace Engineering

Expected May 2029
Cumulative GPA: 3.72/4.0

Experience

UAS Engineering Intern, NASA Ames (AMA) – Moffett Field, CA Jun 2026 – Aug 2026

- Developed autonomous PX4 SITL recovery software in Python/Gazebo/MAVLink for an Air National Guard SAR UAS, using IMU freefall detection, Offboard thrust control, PID vertical-velocity recovery, and Hold handoff
- Analyzed real Black Hawk IMU data to characterize vibration and acceleration, using measured flight behavior to tune deployment-detection thresholds and reduce false-trigger risk in simulation
- Built a Python/PyULog flight-data analysis app for state, vibration, GNSS, and actuator data; used it across 3 prototypes to diagnose anomalies and guide controller, hardware, and configuration changes
- Integrated and revalidated PX4 avionics across 3 flight-ready UAS builds in 3 weeks, configuring motors/ESCs, IMU/magnetometer, GNSS, telemetry, DShot, and battery sensing while resolving flight-system issues
- Planned and executed 10+ flight tests across 4 CG configurations and 3 carrier-UAS releases from 30 ft AGL, authoring test cards/readiness criteria and achieving 3/3 successful manual recoveries
- Led dynamic-stability testing in a 15+ run, 18-hour wind-tunnel campaign from 10–85 ft/s, operating a 4,000 Hz six-axis load-cell DAQ and analyzing PX4/PyULog stability metrics

Test and Visualization Engineering Intern, MeshMap Labs Inc. – San Francisco, CA May 2025 – Sep 2025

- Led UAV/RTK field testing across 4 sites, varying altitude, image overlap, and capture parameters to verify collection performance and establish repeatable operating procedures under changing conditions
- Diagnosed coverage and data-quality failures through post-flight analysis, planned corrective missions, and verified revised acquisition workflows in georeferenced UAV/LiDAR outputs

Full Stack Developer Intern, Axyz Technologies – Oakland, CA Jun 2023 – Jul 2025

- Developed and tested 12+ Python/Django and JavaScript web features across database-backed customer workflows, reproducing defects, validating fixes, and iterating from stakeholder requirements

Projects

Fault-Tolerant Navigation Bench, Personal Project In Progress

- Developing a GNSS/IMU validation bench with Python fault injection for GPS loss, sensor bias/noise, frozen telemetry, and signal dropouts to characterize detection and recovery behavior

Autonomous UAS Systems Research, SDSU Undergraduate Research Mar 2025 – Present

- Integrating sensors, telemetry links, and ground-station controls on an experimental UAV while documenting subsystem interfaces and validating communications ahead of hardware and flight testing

Skills

Software & Simulation: Python, MATLAB, PX4 SITL, Gazebo, MAVLink, PyULog

Flight Systems & Controls: PX4, Offboard Control, PID Control, IMU, GNSS, Magnetometer, Telemetry, Motors/ESCs, DShot

Test & Analysis: Flight Testing, Test Cards, Post-Flight Data Analysis, ULog Analysis, Wind-Tunnel Testing, Six-Axis Load-Cell DAQ